



SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED



LOCATION: NLNG STORAGE TANK	AREA: B
PERMIT TO WORK No.: -	TYPE OF SCAFFOLD: Composite (Independent, independent return & Cantilever)
SCAFFOLD DRAWING No: PMS-WA-ARCO-036	
RISK CLASSIFICATION: ☒ High ☐ Medium ☐ Low	

REFERENCE STANDARDS

- BS EN 1991-1-4:2005 - Wind Actions
- BS EN 1993-1-1:2005 - Design of Steel Structures (Eurocode 3)
- BS EN 12811-1:2003 - Temporary Works Equipment - Scaffolds

SCAFFOLD TO BE INSPECTED & TAGGED BEFORE USE

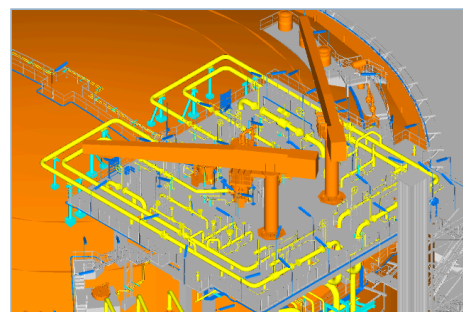
BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of a Composite (Independent, independent return & Cantilever) scaffold required for overhead crane removal and replacement activities at NLNG STORAGE TANK / B. The scaffold envelope is 17m x 3.5m x 7m (length x width x height) and is configured with standards, ledgers, transoms, bracing, platforms, access, and support/tie arrangements as indicated on the approved drawing. The structure is 40m above ground level.

SAFE WORKING LOAD (SWL): 1.5 kN/m²

Note: This is a category 1 engineered scaffold. It must be inspected by the assurance team before use.

SITE / EQUIPMENT PHOTO



TOTAL HORIZONTAL LOAD SUMMARY

Direction	Wind (kN)	Handrail (kN)	TOTAL (kN)
X	43.49	8.7	52.19
Z	63.36	28.95	92.31

Role	ID	Name	Signature	Date
DESIGNED BY	PMS/C3-1141	Nnamani, Ifeanyi		
VERIFIED BY	PMS/612C	Olowo, Peter		
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ERMT NO: PIC/PMS/26/2513

	Project Name:	SCAFFOLD DESIGN REPORT		
	Scaffold Type:	Composite (Independent, independent return & Cantilever)	Calculation Sheet	
	Document Number:	PMS-WA-ARCO-036	Prepared By: Nnamani, Ifeanyi	

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GENERAL NOTE

BS EN 12811-1, 4.2.1.2 Scaffold materials are in accordance with BS EN 12811 with minimum design strength of 210 N/mm². This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 - Design of Steel Structures).

Material Properties

48.3 mm outside diameter x 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

Modelling Philosophy

BS EN 12811-1, 4.2.1.2 A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint ($K_{FX}/K_{FZ} = 20.078$ kN). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot. See [Appendix A](#) for model views.

LOAD CASES

Load diagrams are shown in [Appendix B](#).

Dead Load / Self-weight Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Live Load on Platforms

BS EN 12811-1 Platform live loading was applied to the scaffold working platforms as member UDLs in the STAAD model. The platform load intensity is multiplied by the tributary width to obtain the line load applied to platform supporting members. (LC 2).

Load Class	Load Intensity (kN/m ²)	Tributary Width (m)	Applied STAAD Line Load (kN/m)	Working
2	1.5	0.5	0.75	1.5 x 0.5 = 0.75

Governing platform live load intensity used for SWL = **1.5 kN/m²**.
Total applied platform live load from STAAD output = **104.25 kN**.

Handrail Load

BS EN 12811-1 Handrail loads were applied where required by the scaffold configuration and modelled independently in the global X and Z directions.

Handrail Load (HLX) = **8.7 kN** (member UDL 0.15 kN/m)
Handrail Load (HLZ) = **28.95 kN** (member UDL 0.15 kN/m)

Wind Load

EN 1991-1-4:2005 Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.089452 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

Total Wind Load - X Axis (WLX) = **43.49 kN**
 Total Wind Load - Z Axis (WLZ) = **63.36 kN**

WIND LOAD CALCULATION - EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	V3 =	36.0	m/s
	<i>Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity</i>		
Dredger P. - How to Approximate	Convert 3-second gust to 10-minute mean wind velocity		
	V3 / V3600 =	1.52	
	V600 / V3600 =	1.06	
	V600 = (V600/V3600) x (V3600/V3) x V3 =	25.11	m/s
	Basic wind speed $V_b = V600 =$	25.11	m/s
EN 1991-1-4:2005	Roughness Length (z0) - Terrain Category 0	0.003	m
	Terrain category II (z0,ii)	0.05	
EN 1991-1-4, Table 4.1	Z _{min}	1.0	m
	z - height of structure under consideration	47.0	m
EN 1991-1-4, Sec 4.3	C _{r0} (z) - Roughness factor	1.0	
EN 1991-1-4, Sec 4.3	C _o (z) - Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor K _r = 0.19 x [z0/z0,ii] ^{0.07}	0.156	
EN 1991-1-4, Table 4.1	Z _{min} < z -> Case 1 applies		
	Wind Turbulence		
	C _r (z) = K _r x ln(z/z0)	1.5072	
	Mean velocity V _m (z) = C _r (z) x C _o (z) x V _b	37.839	m/s
EN 1991-1-4, Sec 4.4	K1	1.0	
	Turbulence intensity I _v (z) = k1 / [C _o (z) x ln(z/z0)]	0.1035	
	Peak Velocity Pressure		
	Density of air rho	1.25	kg/m³
EN 1991-1-4, Sec 4.5	q _p (z) = [1 + 7*I _v (z)] x 0.5 x rho x V _m (z)²	1543.3332	N/m²
Wind Load = C_f x q_p x A_f			
C _f =		1.2	
q _p =		1.543333	kN/m²
A _f =	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		0.089452	kN/m

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use gamma = **1.5**;
 SLS (characteristic) combinations use gamma = **1.0**.

DL = Dead Load | **LL** = Platform Live Load | **HLX/HLZ** = Handrail X/Z | **WLX/WLZ** = Wind X/Z

LC No.	Combination	Component Loadings & Factors
7	DL + LL	LC1 x 1.0 + LC2 x 1.0
8	DL + HLX	LC1 x 1.0 + LC3 x 1.0
9	DL + HLZ	LC1 x 1.0 + LC4 x 1.0

LC No.	Combination	Component Loadings & Factors
10	DL + WLX	LC1 x 1.0 + LC5 x 1.0
11	DL + WLZ	LC1 x 1.0 + LC6 x 1.0
12	DL + LL + HLX	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0
13	DL + LL + HLZ	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0
14	DL + LL + WLX	LC1 x 1.0 + LC2 x 1.0 + LC5 x 1.0
15	DL + LL + WLZ	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0
16	DL + LL + HLX + WLX	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0 + LC5 x 1.0
17	DL + LL + HLZ + WLZ	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0 + LC6 x 1.0
18	DL + LL + HLX + WLZ	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0 + LC6 x 1.0
19	DL + LL + HLZ + WLX	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0 + LC5 x 1.0
20	1.5DL	LC1 x 1.5
21	1.5DL + 1.5LL	LC1 x 1.5 + LC2 x 1.5
22	1.5DL + 1.5HLX	LC1 x 1.5 + LC3 x 1.5
23	1.5DL + 1.5HLZ	LC1 x 1.5 + LC4 x 1.5
24	1.5DL + 1.5WLX	LC1 x 1.5 + LC5 x 1.5
25	1.5DL + 1.5WLZ	LC1 x 1.5 + LC6 x 1.5
26	1.5DL + 1.5LL + 1.5HLX	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5
27	1.5DL + 1.5LL + 1.5HLZ	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5
28	1.5DL + 1.5LL + 1.5WLX	LC1 x 1.5 + LC2 x 1.5 + LC5 x 1.5
29	1.5DL + 1.5LL + 1.5WLZ	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5
30	1.5DL + 1.5LL + 1.5HLX + 1.5WLX	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5 + LC5 x 1.5
31	1.5DL + 1.5LL + 1.5HLZ + 1.5WLZ	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5 + LC6 x 1.5
32	1.5DL + 1.5LL + 1.5HLX + 1.5WLZ	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5 + LC6 x 1.5
33	1.5DL + 1.5LL + 1.5HLZ + 1.5WLX	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5 + LC5 x 1.5

STATIC CHECK PARAMETERS - UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.773 at Member No. **402**, Load Combination **31**, Clause **EC-6.3.3-662**. **PASS** - All members adequate; UC ratio below unity.

UC ratio diagram: see [Appendix C](#).

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
402	PIP483.6	PASS	EC-6.3.3-662	0.773	31	2.82	C
355	PIP483.6	PASS	EC-6.2.9.1	0.754	31	1.91	T
449	PIP483.6	PASS	EC-6.2.9.1	0.751	31	1.94	T
1055	PIP483.6	PASS	EC-6.2.7(5)	0.750	31	0.29	C
1041	PIP483.6	PASS	EC-6.2.7(5)	0.748	31	0.25	C
1039	PIP483.6	PASS	EC-6.2.7(5)	0.741	31	0.30	C
1042	PIP483.6	PASS	EC-6.2.7(5)	0.741	31	0.24	C
308	PIP483.6	PASS	EC-6.3.3-662	0.735	31	2.92	C
1057	PIP483.6	PASS	EC-6.2.7(5)	0.735	31	0.23	C
1040	PIP483.6	PASS	EC-6.2.7(5)	0.732	31	0.30	C
496	PIP483.6	PASS	EC-6.3.3-662	0.728	31	2.62	C
1058	PIP483.6	PASS	EC-6.2.7(5)	0.712	31	0.26	C
1045	PIP483.6	PASS	EC-6.2.9.1	0.675	31	0.29	C
1052	PIP483.6	PASS	EC-6.2.9.1	0.675	31	0.28	C
1043	PIP483.6	PASS	EC-6.2.9.1	0.673	31	0.28	C
1044	PIP483.6	PASS	EC-6.2.9.1	0.672	31	0.28	C

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
1048	PIP483.6	PASS	EC-6.2.9.1	0.672	31	0.31	C
1050	PIP483.6	PASS	EC-6.2.9.1	0.672	31	0.29	C
1051	PIP483.6	PASS	EC-6.2.9.1	0.672	31	0.28	C
1053	PIP483.6	PASS	EC-6.2.9.1	0.672	31	0.29	C
979	PIP483.6	PASS	EC-6.3.3-662	0.671	31	2.62	C
1049	PIP483.6	PASS	EC-6.2.9.1	0.671	31	0.29	C
978	PIP483.6	PASS	EC-6.3.3-662	0.670	31	3.10	C
977	PIP483.6	PASS	EC-6.3.3-662	0.668	31	2.61	C
1046	PIP483.6	PASS	EC-6.2.9.1	0.668	31	0.30	C
1047	PIP483.6	PASS	EC-6.2.9.1	0.668	31	0.31	C
1054	PIP483.6	PASS	EC-6.2.9.1	0.663	31	0.30	C
356	PIP483.6	PASS	EC-6.3.3-662	0.662	31	3.61	C
450	PIP483.6	PASS	EC-6.3.3-662	0.660	31	3.63	C
980	PIP483.6	PASS	EC-6.3.3-662	0.654	31	3.17	C
1056	PIP483.6	PASS	EC-6.2.9.1	0.643	31	0.28	C
976	PIP483.6	PASS	EC-6.3.3-662	0.642	31	2.73	C
403	PIP483.6	PASS	EC-6.2.9.1	0.636	31	0.38	T
160	PIP483.6	PASS	EC-6.2.7(5)	0.629	31	2.17	T
981	PIP483.6	PASS	EC-6.3.3-662	0.621	31	2.71	C
497	PIP483.6	PASS	EC-6.2.9.1	0.604	31	0.48	T
309	PIP483.6	PASS	EC-6.2.9.1	0.601	31	0.18	T
1023	PIP483.6	PASS	EC-6.2.7(5)	0.592	31	0.21	C
543	PIP483.6	PASS	EC-6.2.7(5)	0.582	31	4.49	T
1059	PIP483.6	PASS	EC-6.2.7(5)	0.569	31	0.62	C
957	PIP483.6	PASS	EC-6.3.3-662	0.554	31	2.06	C
1060	PIP483.6	PASS	EC-6.2.7(5)	0.552	31	0.97	C
448	PIP483.6	PASS	EC-6.3.3-662	0.526	32	1.16	C
965	PIP483.6	PASS	EC-6.3.3-662	0.526	31	4.04	C
354	PIP483.6	PASS	EC-6.3.3-662	0.523	32	1.13	C
1022	PIP483.6	PASS	EC-6.2.9.1	0.516	31	0.22	C
222	PIP483.6	PASS	EC-6.3.3-662	0.512	31	6.70	C
401	PIP483.6	PASS	EC-6.2.9.1	0.509	32	1.07	T
544	PIP483.6	PASS	EC-6.3.3-662	0.505	31	6.14	C
495	PIP483.6	PASS	EC-6.2.9.1	0.501	32	1.18	T
Showing highest 50 utilization ratios; 959 lower-utilization members omitted.							

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **4.114 kN** (Member 541, C). The required right-angle coupler class is selected from Table C.1 below using the slipping resistance limit.

Selected coupler class: **Class A** (capacity = 10.0 kN).

Check: 4.114 kN ≤ 10.0 kN -> **PASS**

Class A right-angle couplers are adequate.

Member force table: see [Appendix D](#).

Table C.1 - Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force $F_{s,k}$ (kN)	10.0	15.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.8
	Pull-apart force $F_{p,k}$ (kN)	20.0	30.0
	Rotational moment $M_{T,k}$ (kNm)	-	0.13
Swivel Coupler (SW)	Slipping force $F_{s,k}$ (kN)	6.0	9.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.8
	Pull-apart force $F_{p,k}$ (kN)	15.0	22.5
	Rotational moment $M_{T,k}$ (kNm)	-	0.08
Putlog / Single Coupler (PL)	Slipping force $F_{s,k}$ (kN)	-	3.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.6

FRICTIONAL RESISTANCE

Max vertical reaction per support ($\text{Max } f_y$) = Resistance x 0.15 / C_F	= 10.039 kN
Coefficient of friction (C_F)	= 0.3
Frictional Resistance (FR) = C_F x Max f_y	= 3.0117 kN
Spring Stiffness = FR / 0.15	= 20.078 kN <- applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state - values from STAAD.Pro SLS (unfactored) combination results. Diagrams and displacement tables are in [Appendix C](#) and [Appendix D](#).

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement - LC 17 Member 902 Member length L = 1800.0 mm	1.561 mm	$L/100 = 1800/100 = 18.0 \text{ mm}$	PASS

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement - LC 17 Member 978 Member length L = 1000.0 mm	1.295 mm	$L/200 = 1000/200 = 5.0 \text{ mm}$	PASS

CONCLUSION

The structural analysis and design of the **Composite (Independent, independent return & Cantilever)** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = 0.773 at Member 402). Vertical and horizontal deflections are within their respective allowable limits of $L/100 = 18.0\text{mm}$ and $L/200 = 5.0\text{mm}$. All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and

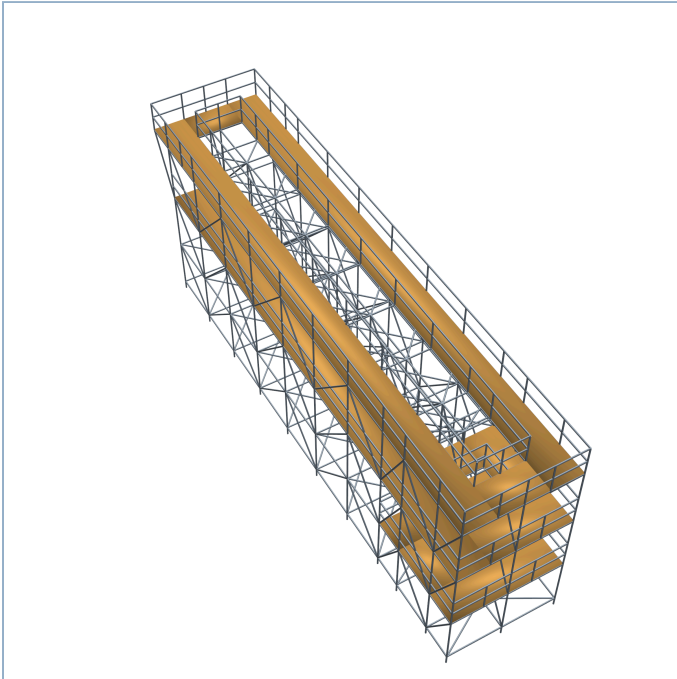
material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

INSTALLATION NOTES

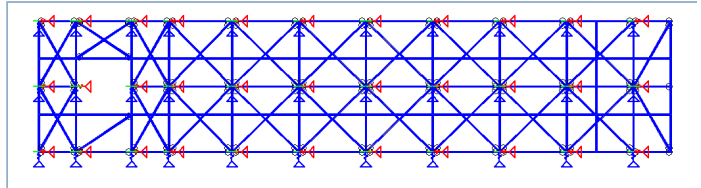
1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. Confirm that the Composite (Independent, independent return & Cantilever) scaffold configuration matches the approved drawing and is suitable for overhead crane removal and replacement activities.
3. The work vicinity must be barricaded and appropriate signage placed at all access points.
4. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
5. A valid work permit must be obtained, discussed, and signed before erection commences.
6. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
7. Install and secure cantilever support members, needles, anchors, ties, and back-span restraints exactly as shown on the approved drawing before building the working platform.
8. Build the scaffold frame to the approved envelope of 17.0m x 3.5m x 7.0m, checking level, plumb, and cantilever projection at each lift.
9. Install bracing, ledgers, transoms, platform boards, guardrails, and toe boards progressively, maintaining all specified ties and restraints.
10. Install mid-level horizontal ledgers and transoms at 2.0m height in accordance with the working drawing.
11. Install mid-level horizontal ledgers and transoms at 2.5m height in accordance with the working drawing.
12. Install mid-level horizontal ledgers and transoms at 3.0m height in accordance with the working drawing.
13. Install mid-level horizontal ledgers and transoms at 4.0m height in accordance with the working drawing.
14. Install mid-level horizontal ledgers and transoms at 4.5m height in accordance with the working drawing.
15. Install mid-level horizontal ledgers and transoms at 5.0m height in accordance with the working drawing.
16. Install mid-level horizontal ledgers and transoms at 6.0m height in accordance with the working drawing.
17. Install mid-level horizontal ledgers and transoms at 6.5m height in accordance with the working drawing.
18. Install all plan, face, and longitudinal bracing shown on the approved drawing to provide sway resistance in both principal directions.
19. Install platform boards, guardrails, mid-rails, toe boards, access arrangements, and any handrail load-resisting members required by the design.
20. Tighten and torque-check all load-bearing couplers and support connections before the scaffold is released for inspection.
21. Gravlock or box-tie the scaffold structure to the existing structure at all indicated tie points at 2.0m and 6.0m height as shown on the working drawing.
22. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before use.
23. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
24. All modifications made after erection must be carried out by a competent scaffold team with the approval of the scaffold engineer.
End-user or third-party modification of this scaffold structure is strictly prohibited.
25. For dismantling, perform the erection steps in reverse order.

APPENDIX A - MODEL VIEWS

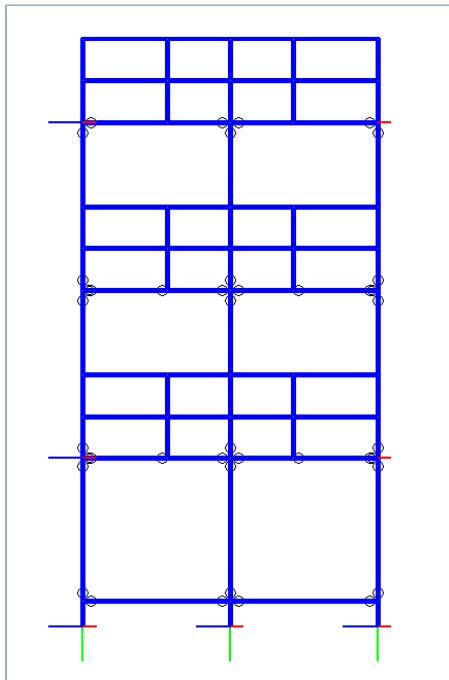
3D Isometric View



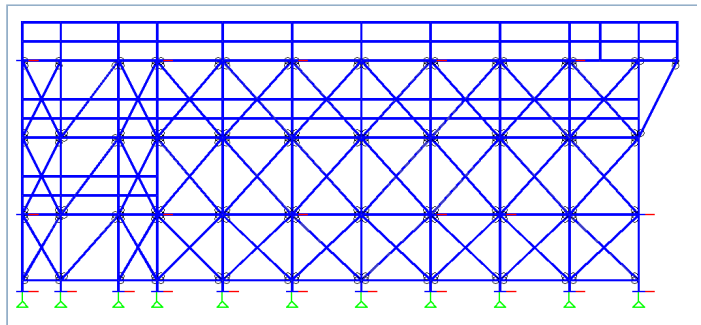
Plan View



Elevation from +X Axis



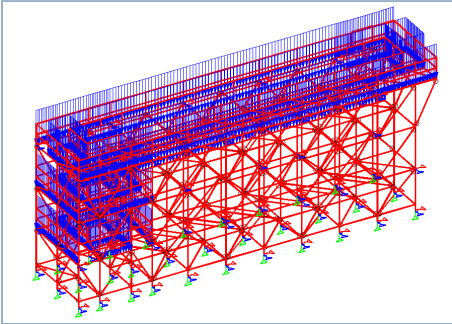
Elevation from +Z Axis



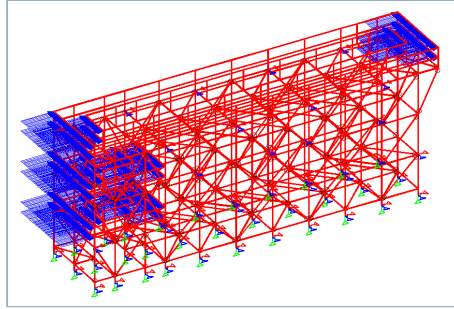
Scaffold Dimensions: W = 17.0m | D = 3.5m | H = 7.0m | **Lifts:** 0.3m, 2.0m, 2.5m, 3.0m, 4.0m, 4.5m, 5.0m, 6.0m, 6.5m, 7.0m

APPENDIX B - LOAD DIAGRAMS

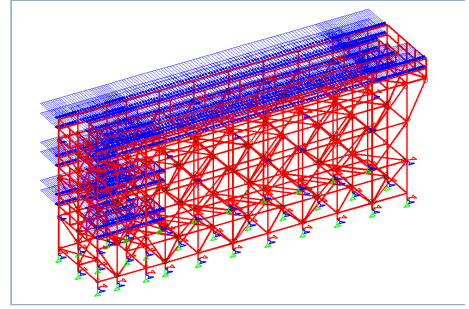
Live Load - Platform Live Load (LL)



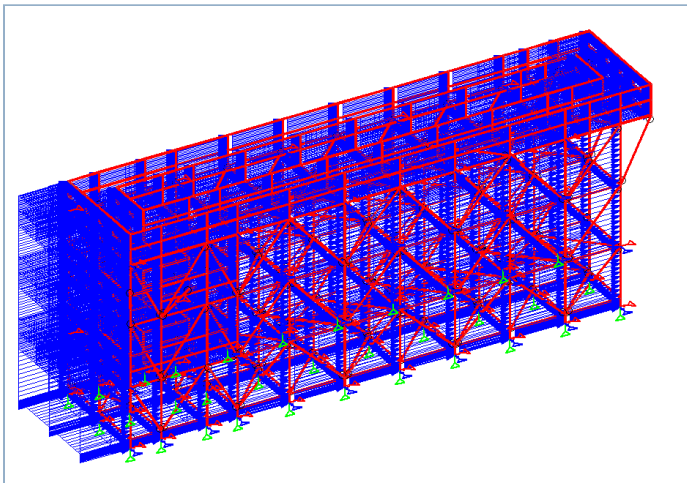
Handrail Load in X Direction (HLX)



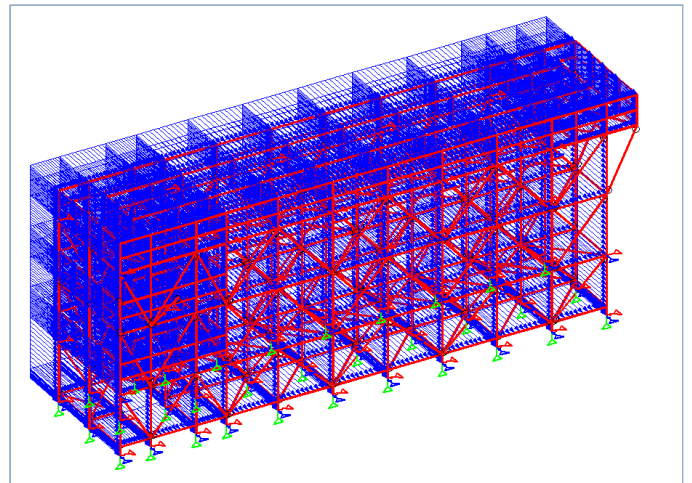
Handrail Load in Z Direction (HLZ)



Wind Load - X Axis (WLX)

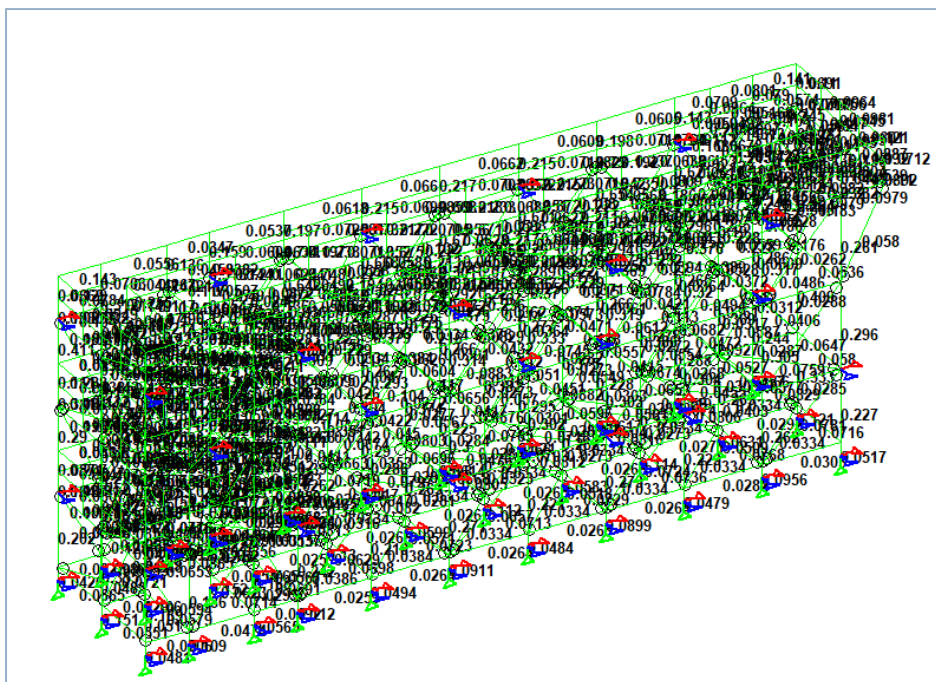


Wind Load - Z Axis (WLZ)

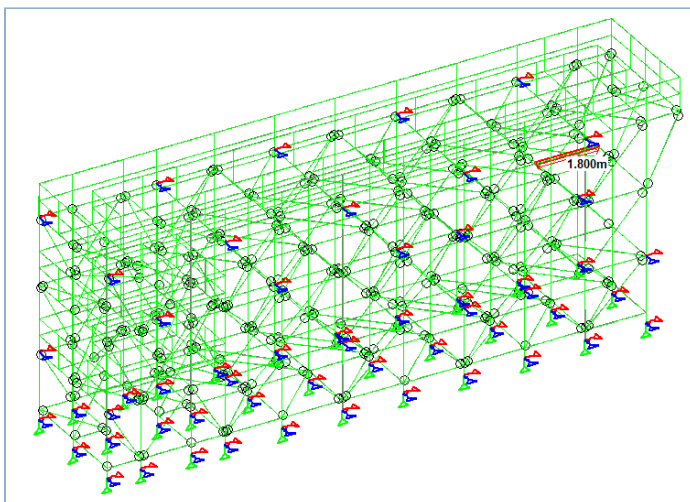


APPENDIX C - ANALYSIS RESULTS

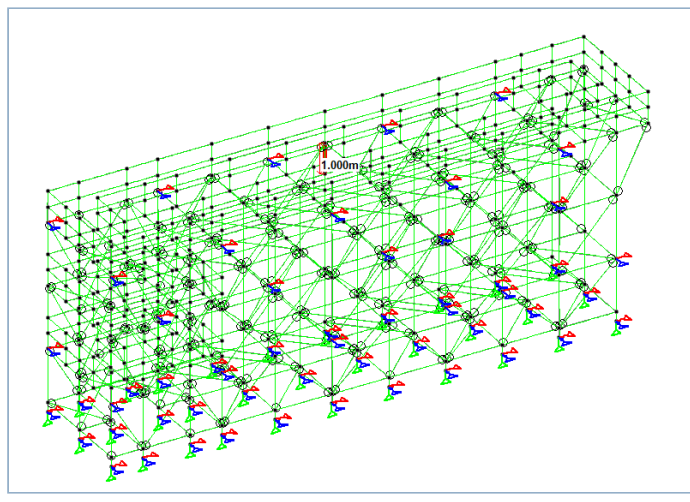
Utilization Ratio (UC) Diagram



Vertical Deflection Diagram



Horizontal Deflection Diagram



APPENDIX D - GUI RESULTS TABLES

Member Force / Code Check Table (Connection Stability)

Beam	L/C	Node	Axial Force kN	Shear-Y kN	Shear-Z kN	Torsion kN-m	Moment-Y kN-m	Moment-Z kN-m
541	17	255	4.114	-0.085	-0.049	0.007	-0.022	0.082
544	17	256	4.094	-0.742	0.031	0.004	-0.011	-0.455
824	17	273	3.170	0.493	-0.093	-0.051	-0.020	-0.108
561	17	372	3.123	0.116	0.184	0.021	0.084	0.025
151	17	77	3.098	0.298	-0.036	-0.000	-0.012	0.012
168	17	84	3.084	-0.081	-0.015	0.022	0.005	-0.059
542	17	254	3.022	0.663	0.042	-0.027	-0.021	0.408
543	17	254	2.995	1.155	-0.032	-0.034	-0.016	-0.328
144	17	68	2.976	-0.034	0.034	-0.015	-0.021	-0.070
157	17	65	2.966	0.271	-0.017	0.001	-0.015	-0.169
57	17	25	2.935	0.314	-0.029	0.006	-0.018	-0.021
44	17	28	2.857	0.659	0.031	-0.014	-0.018	0.172
541	11	255	2.623	0.025	-0.026	0.002	-0.011	-0.024
561	19	372	2.622	0.120	0.146	0.005	0.059	0.022
544	11	256	2.618	-0.175	0.022	-0.000	-0.009	-0.131
541	18	255	2.604	-0.281	-0.050	0.007	-0.025	0.181
541	15	255	2.604	-0.290	-0.032	0.007	-0.015	0.185
561	18	372	2.590	0.100	0.104	0.014	0.051	0.032
544	18	256	2.584	-0.472	-0.003	0.005	0.005	-0.348
544	15	256	2.583	-0.481	0.015	0.005	-0.004	-0.352
824	18	273	2.582	0.532	-0.072	-0.031	-0.015	-0.127
824	19	273	2.478	0.531	0.017	-0.032	0.014	-0.125
447	17	213	2.424	-0.039	-0.006	0.001	-0.004	0.079
450	17	214	2.422	-1.040	-0.002	-0.000	0.001	-0.637
353	17	171	2.410	-0.034	-0.002	0.000	-0.001	0.078
356	17	172	2.407	-1.044	0.003	0.000	-0.001	-0.640
541	17	255	2.387	0.388	0.387	0.017	0.001	0.000

Vertical Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
902	1.561	0.900	17	1153	-0.140	-3.713	5.943
902	1.559	0.900	19	1154	0.077	-3.642	1.320
902	1.557	0.900	15	1155	-0.134	-3.618	4.683
902	1.557	0.900	13	1155	-0.023	-3.640	1.329
902	1.556	0.900	18	1156	-0.120	-3.616	4.688
902	1.555	0.900	14	1157	0.083	-3.548	0.060
902	1.554	0.900	16	1158	0.097	-3.547	0.065
902	1.554	0.900	7	1158	-0.016	-3.547	0.069
902	1.553	0.900	12	1158	-0.004	-3.546	0.074
556	1.078	0.900	17	1670	0.022	-1.214	6.028
556	1.073	0.900	15	1677	0.019	-1.227	4.723
556	1.072	0.900	18	1679	0.021	-1.227	4.730
556	1.068	0.900	19	1685	0.030	-1.223	1.385
556	1.067	0.900	14	1687	0.027	-1.236	0.081
556	1.066	0.900	16	1688	0.029	-1.235	0.087
556	1.066	0.900	13	1689	0.008	-1.223	1.376
556	1.065	0.900	7	1689	0.005	-1.236	0.071
556	1.064	0.900	12	1690	0.007	-1.235	0.078
517	0.951	0.900	17	1893	0.052	-3.489	0.149
892	0.948	0.900	7	1898	-0.019	-3.410	0.077
517	0.945	0.900	18	1904	0.102	-3.121	0.126
517	0.945	0.900	15	1904	0.040	-3.127	0.125
517	0.942	0.900	19	1911	0.280	-3.291	0.028
517	0.941	0.900	13	1912	0.043	-3.314	0.022
518	0.940	0.900	16	1914	0.336	-3.190	-0.006
518	0.940	0.900	14	1914	0.272	-3.196	-0.004

Max = 1.561 mm (LC 17) | L = Member 902 length (1800 mm) | Allowable L/100 = 18.0 mm -> **PASS**

Horizontal Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
978	1.295	0.583	17	772	0.016	-0.159	0.393
979	1.293	0.583	17	773	0.006	-0.209	0.342
977	1.287	0.583	17	777	0.003	-0.203	0.309
980	1.232	0.583	17	811	0.014	-0.161	0.542
976	1.216	0.583	17	822	0.012	-0.144	0.453
533	1.203	1.500	17	1662	-0.008	-0.118	5.592
486	1.155	1.500	17	1732	-0.005	-0.162	5.513
439	1.138	1.500	17	1757	0.000	-0.105	5.455
392	1.123	1.500	17	1780	0.000	-0.161	5.393
345	1.120	1.500	17	1785	0.003	-0.110	5.312
981	1.093	0.667	17	914	0.051	-0.202	0.371
533	1.088	1.500	18	1838	-0.006	-0.118	4.530
533	1.087	1.500	15	1839	-0.008	-0.118	4.526
290	1.068	1.500	17	1872	-0.003	-0.181	5.176
398	1.066	1.167	14	1877	0.136	-0.106	1.088
486	1.065	1.500	18	1878	-0.004	-0.161	4.477
486	1.064	1.500	15	1879	-0.006	-0.162	4.473
398	1.064	1.167	16	1879	0.122	-0.106	1.091
957	1.064	0.583	17	939	-0.045	-0.160	0.428
398	1.059	1.167	12	1888	-0.018	-0.107	1.092
439	1.059	1.500	18	1889	0.001	-0.104	4.430
398	1.059	1.167	7	1889	-0.003	-0.107	1.089
439	1.058	1.500	15	1890	-0.002	-0.105	4.426
445	1.055	1.167	14	1894	0.130	-0.149	1.077
445	1.055	1.167	16	1896	0.115	-0.149	1.079
445	1.050	1.167	12	1904	-0.017	-0.149	1.083

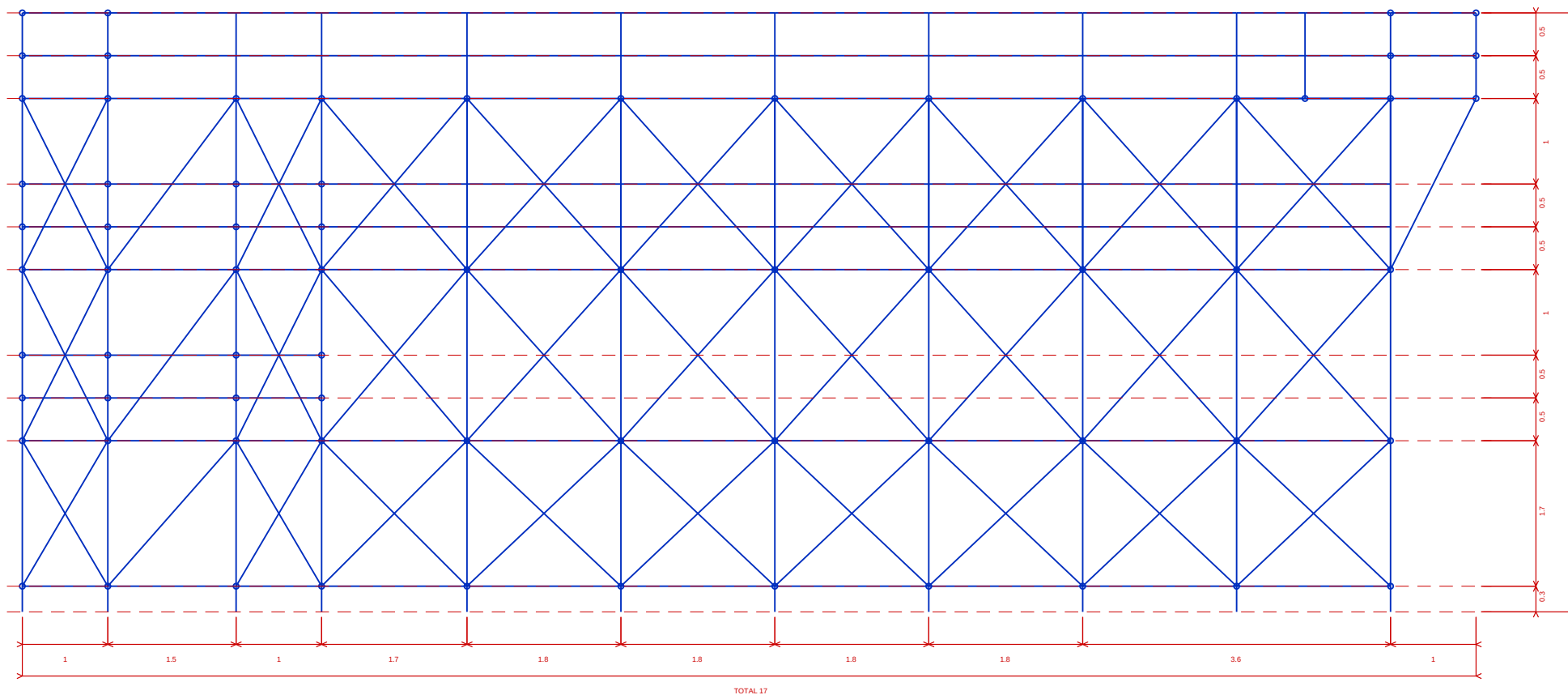
Max = 1.295 mm (LC 17) | L = Member 978 length (1000 mm) | Allowable L/200 = 5.0 mm -> **PASS**

APPENDIX E - REFERENCES & STANDARDS

Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment - Scaffolds - Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures - General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures - Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Dredger P.	How to Approximate 10-Minute Mean Wind Speed from 3-Second Gust	-
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 - Structural Analysis and Design Software	2022

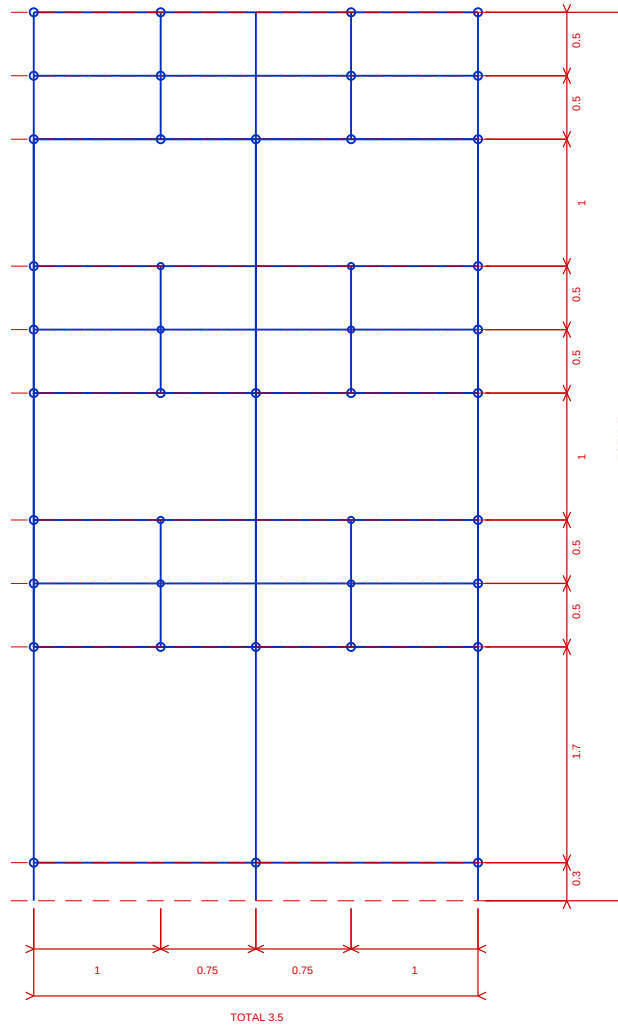
Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) - 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.

Report: Scaffold Structural Design Report - ARCO M&E, NLNG Bonny Island.



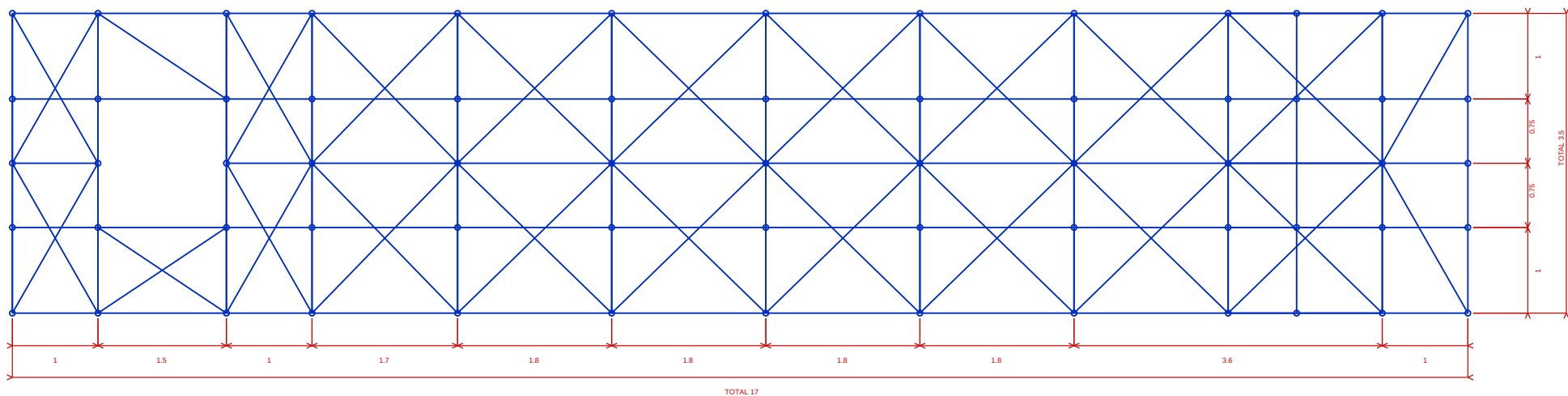
FRONT VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-FRONT-VIEW			 
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - FRONT VIEW		SCALE NTS	DATE 16-JUN-2026	REV 0	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	COMPANY
					SHEET 1 OF 4
					SIZE A4



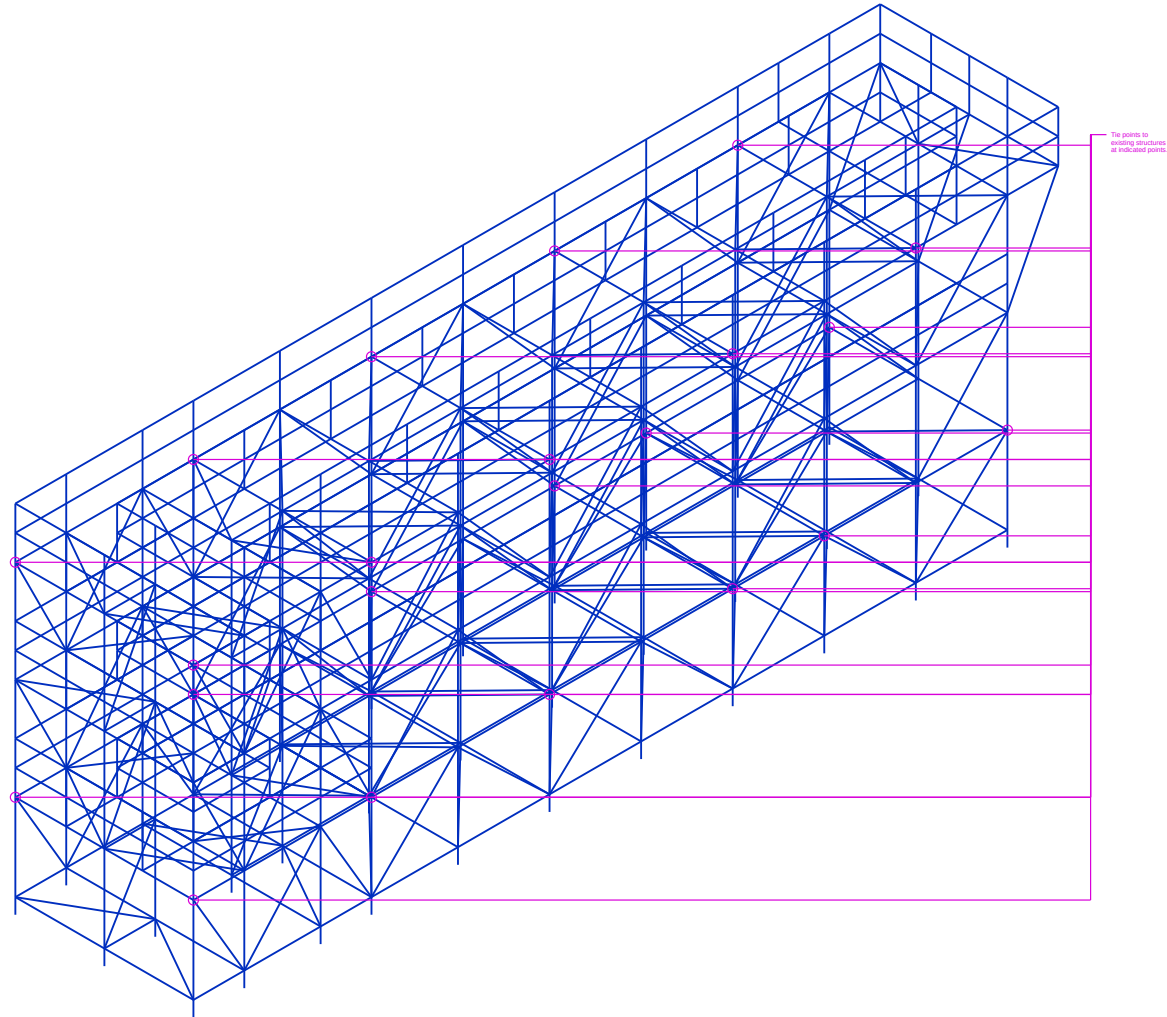
SIDE VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-SIDE-VIEW			 NLNG	
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - SIDE VIEW		SCALE NTS	DATE 16-JUN-2026	REV 0		
CLIENT		LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	COMPANY	
					SHEET 2 OF 4	SIZE A4



PLAN VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-PLAN-VIEW			 
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - PLAN VIEW		SCALE NTS	DATE 16-JUN-2026	REV 0	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	COMPANY
					SHEET 3 OF 4
					SIZE A4



3D VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-3D-VIEW			 
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - 3D VIEW		SCALE NTS	DATE 16-JUN-2026	REV 0	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	COMPANY
					SHEET 4 OF 4
					SIZE A4

3D Isometric View

